

MIT OpenCourseWare

2.006

Thermal-Fluids Engineering II 2.006 — Thermal-Fluids Engineering II May 2026

EXTREMELY HARD – MIT LEVEL**Time Limit: 3 hours****Total Marks: 100**

1. (2 points) Analyze the limitations of standard approaches to Convection when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (5 points) Construct an original proof-level problem involving Radiation for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
3. (4 points) Analyze the limitations of standard approaches to Multiphase Flow when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
4. (6 points) Construct an original proof-level problem involving Combustion for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
5. (2 points) Construct an original proof-level problem involving Turbomachinery for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
6. (5 points) Analyze the limitations of standard approaches to Convection when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (7 points) Prove the fundamental theorem concerning Radiation in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (3 points) Prove the fundamental theorem concerning Multiphase Flow in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (2 points) Analyze the limitations of standard approaches to Combustion when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
10. (3 points) Construct an original proof-level problem involving Turbomachinery for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
11. (5 points) Prove the fundamental theorem concerning Convection in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (5 points) Analyze the limitations of standard approaches to Radiation when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
13. (6 points) Analyze the limitations of standard approaches to Multiphase Flow when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (5 points) Prove the fundamental theorem concerning Combustion in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
15. (4 points) Prove the fundamental theorem concerning Turbomachinery in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (2 points) Analyze the limitations of standard approaches to Convection when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.
17. (3 points) Construct an original proof-level problem involving Radiation for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
18. (2 points) Prove the fundamental theorem concerning Multiphase Flow in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (3 points) Analyze the limitations of standard approaches to Combustion when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (6 points) Prove the fundamental theorem concerning Turbomachinery in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
21. (5 points) Construct an original proof-level problem involving Convection for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (5 points) Prove the fundamental theorem concerning Radiation in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Prove the fundamental theorem concerning Multiphase Flow in the context of 2.006 (Thermal-Fluids Engineering II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (5 points) Construct an original proof-level problem involving Combustion for 2.006 (Thermal-Fluids Engineering II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
25. (3 points) Analyze the limitations of standard approaches to Turbomachinery when applied to edge cases in 2.006. Construct explicit counterexamples showing where intuitive reasoning fails.

Solutions

Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Convection. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 2:

Solution approach: First recall the definitions and key theorems related to Radiation from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 3:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Multiphase Flow. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 4:

Solution approach: First recall the definitions and key theorems related to Combustion from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 5:

Solution approach: First recall the definitions and key theorems related to Turbomachinery from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Convection. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Combustion. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 10:

Solution approach: First recall the definitions and key theorems related to Turbo-machinery from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 11:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Radiation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 13:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Multiphase Flow. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 14:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 15:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 16:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Convection. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 17:

Solution approach: First recall the definitions and key theorems related to Radiation from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 18:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments

of Combustion. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Convection from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 22:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The proof follows from the definitions and properties established in 2.006. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

Solution approach: First recall the definitions and key theorems related to Combustion from 2.006. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 25:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Turbomachinery. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.